SQL Project:- Exploring Customer Behavior

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Batch:-164 Business Analysis

Introduction

In today's competitive electronics retail industry, understanding customer purchasing behavior is essential for improving sales performance and customer satisfaction. This project focuses on analyzing customer transaction data from an electronics store using SQL. The database consists of customer information, product details, sales transactions, and premium membership data.

By analyzing these datasets, we can identify purchasing patterns, high-value customers, top-selling products, revenue trends, and the effectiveness of the Gold Membership Program. The insights generated from this analysis can help businesses make data-driven decisions regarding marketing strategies, inventory management, customer retention, and loyalty programs.

Objective of the Project

The main objectives of this project are:

- To analyze customer purchasing behavior using SQL queries.
- To identify the highest revenue-generating products.
- To determine the most valuable customers based on spending patterns.
- To evaluate the performance of the Gold Membership Program.
- To analyze yearly sales and revenue trends.
- To identify popular products among customers.
- To measure customer engagement through purchase frequency.
- To generate business insights that support decision-making and improve customer retention.

Dataset Description:-

This project uses five datasets:

1. Users

Contains customer registration information including User ID and Signup Date.

2. User_Name

Contains User ID and corresponding customer names.

3. Products

Contains electronic product details such as Product ID, Product Name, and Price.

4. Sales

Contains transaction records including User ID, Product ID, and Purchase Date.

5. GoldUsers_Signup

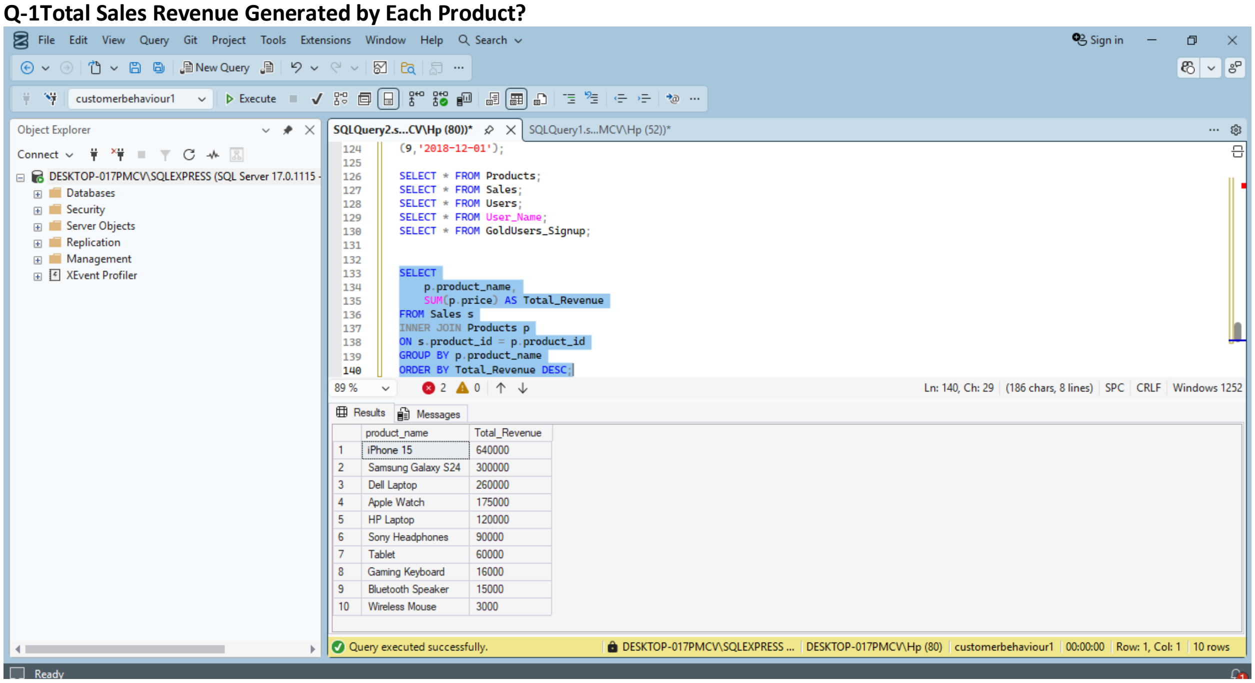
Contains information about customers who subscribed to the Gold Membership Program and their membership start date.

Tools and Technologies Used

- SQL Server Management Studio (SSMS)

- SQL Queries
- Joins
- Aggregate Functions
- Window Functions
- Date Functions
- Data Analysis Techniques

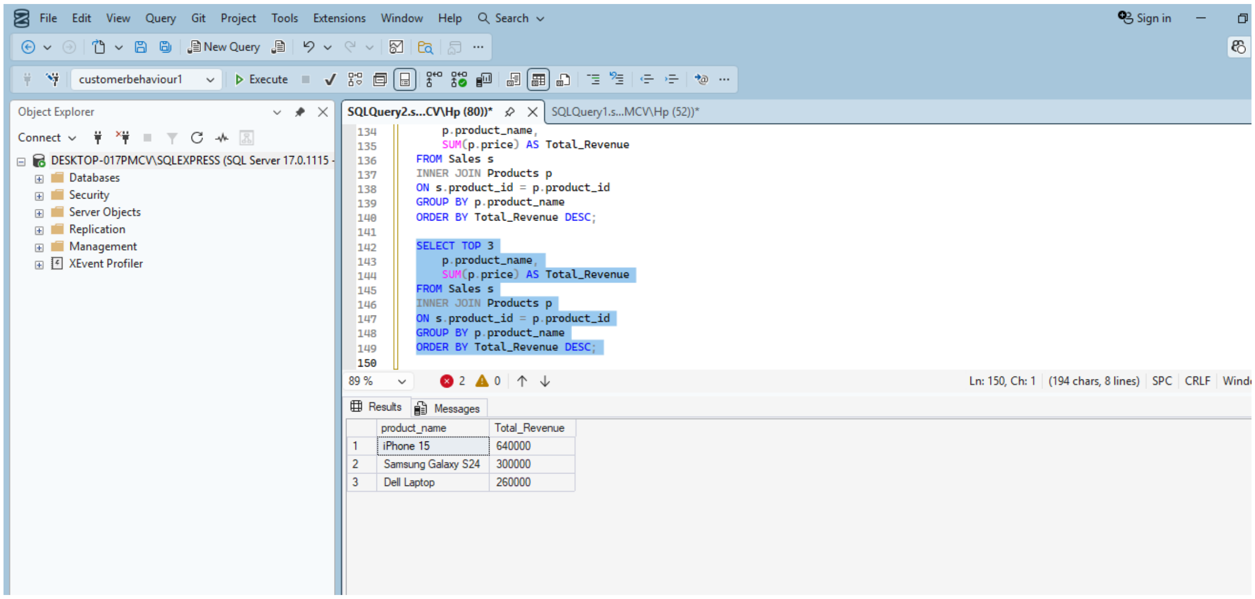
Q-1Total Sales Revenue Generated by Each Product?



Key Findings;-

- The highest revenue was generated by Iphone 15
- Premium electronic products contributed significantly to total sales revenue.
- Products such as wireless mouse generated comparatively lower revenue.
- Revenue distribution indicates that customers prefer high-value electronic products.
- The company should focus marketing efforts on top-performing products while improving sales strategies for lower-performing products.

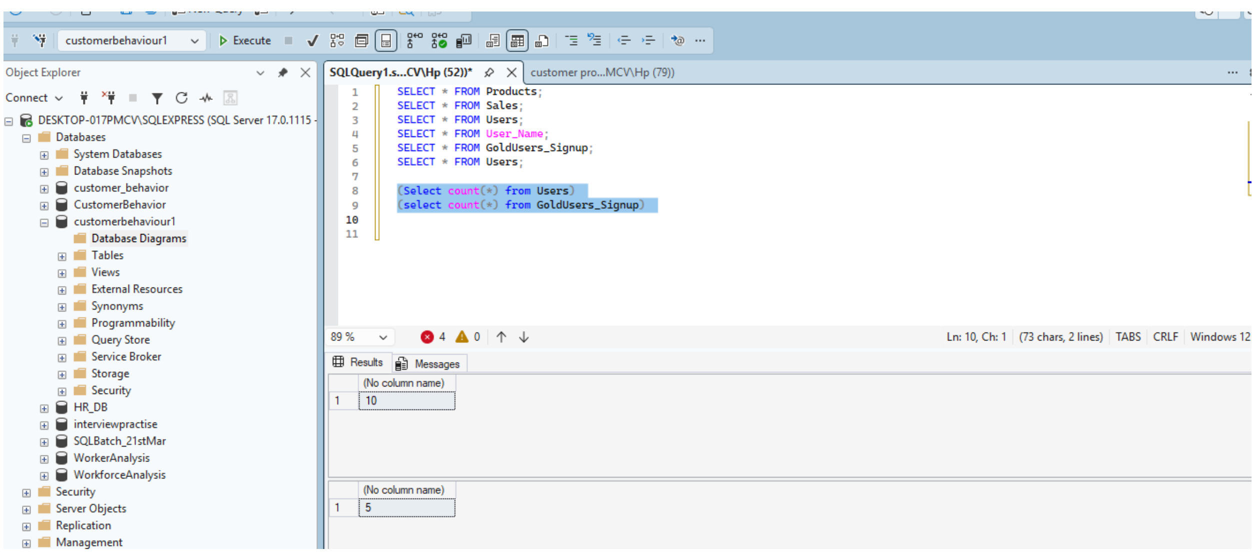
Q2. Which 3 Products Have the Highest Sales Revenue?



Key Findings;-

- The top 3 products i.e., (iPhone 15, Samsung Galaxy S24 and Dell Laptop) contributed the highest share of total revenue.
- These products represent the most profitable product categories in the business.
- The company should ensure sufficient inventory availability for these products.
- Marketing campaigns should prioritize these products to maximize revenue growth.
- These products can be used as flagship offerings to attract more customers.

Q3. How Many Users Have Signed Up for the Service and Taken Gold Membership?



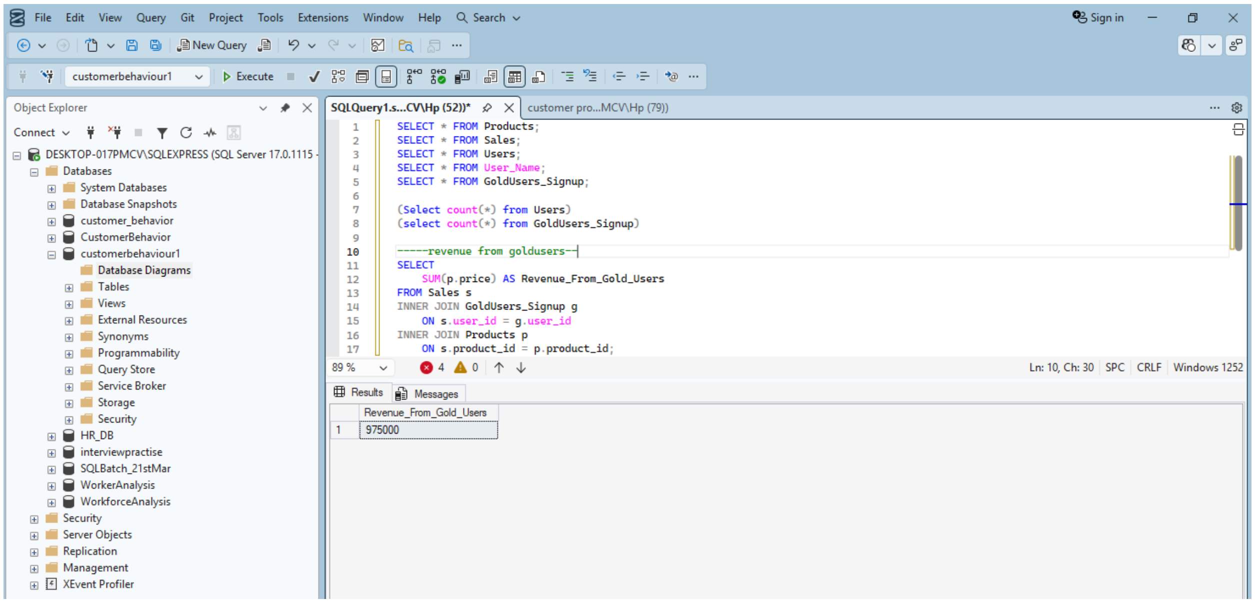
Key Findings;-

- A total of 10 customers registered on the platform.
- Out of these, 5 customers upgraded to Gold Membership.

- The Gold Membership adoption rate is 50%.
- This indicates a good level of customer interest in the premium membership program.
- The business can further improve membership adoption through exclusive offers and loyalty rewards.

Approximately 50% of registered users subscribed to the Gold Membership Program, indicating moderate customer engagement with premium services.

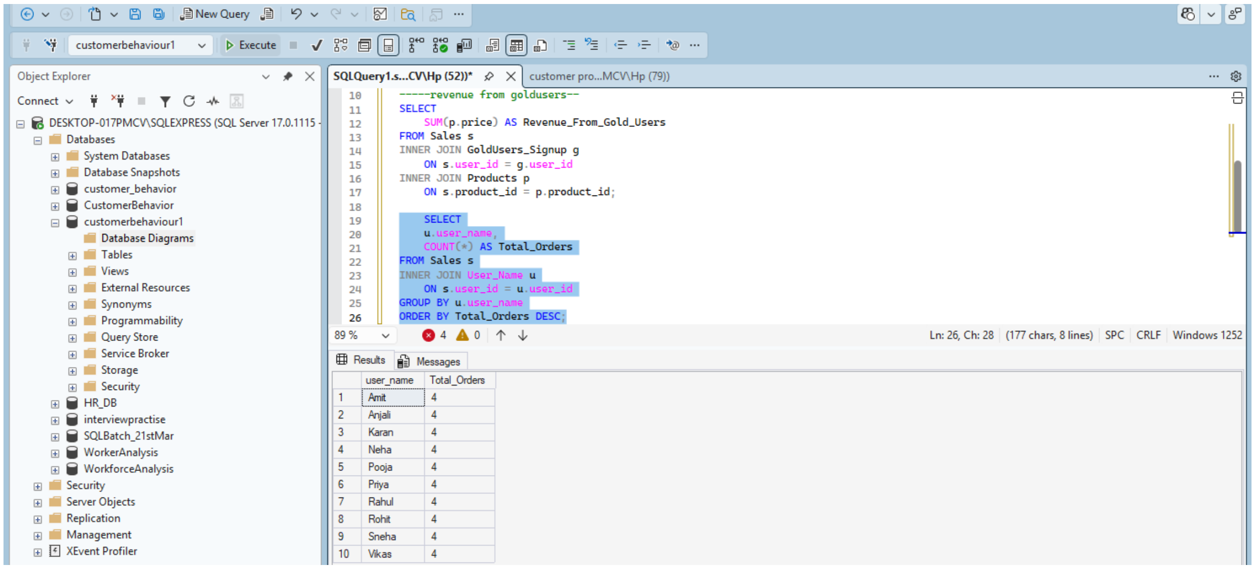
Q4. What is the Revenue Generated from Gold Users?



Key Findings;-

- Gold Members contributed a significant portion of total sales revenue.
- Premium members demonstrated higher purchasing activity compared to regular customers.
- The Gold Membership Program appears to positively influence customer spending behavior.
- Revenue generated by Gold Members highlights the importance of customer loyalty programs.
- The company should continue investing in Gold Membership benefits to increase customer retention and revenue.

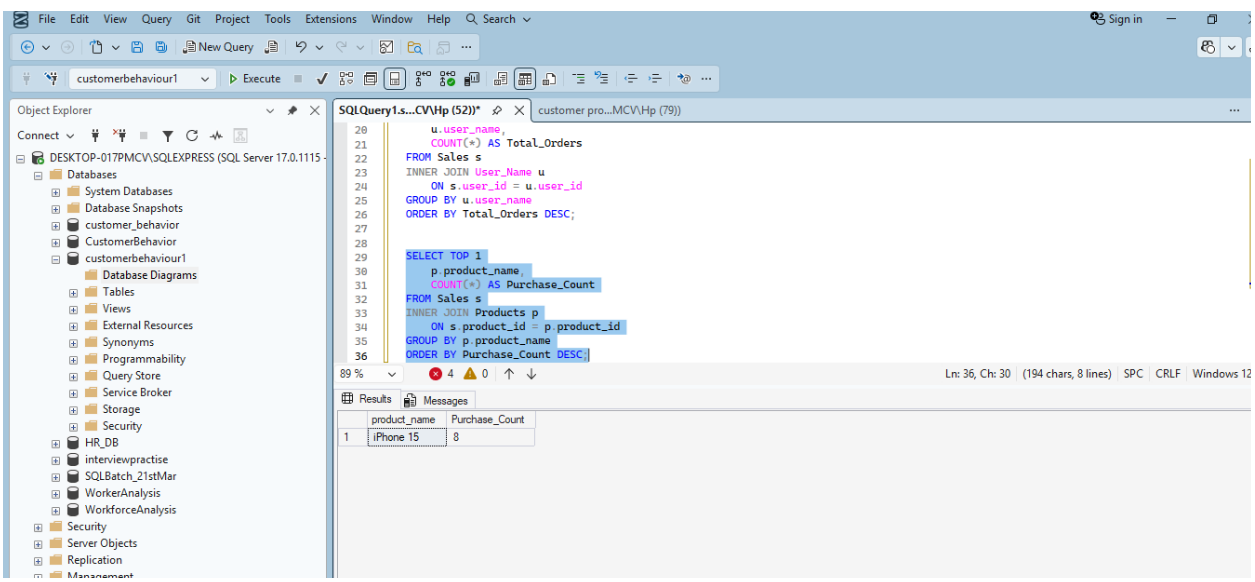
Q5. What is the Frequency of Customer Purchases?



Key Findings;-

- The analysis identified the most active customers on the platform.
- Frequent purchasers demonstrate strong customer engagement.
- Some customers purchase significantly more often than others.
- Highly active customers may be suitable candidates for loyalty rewards.
- Purchase frequency is an important indicator of customer retention.

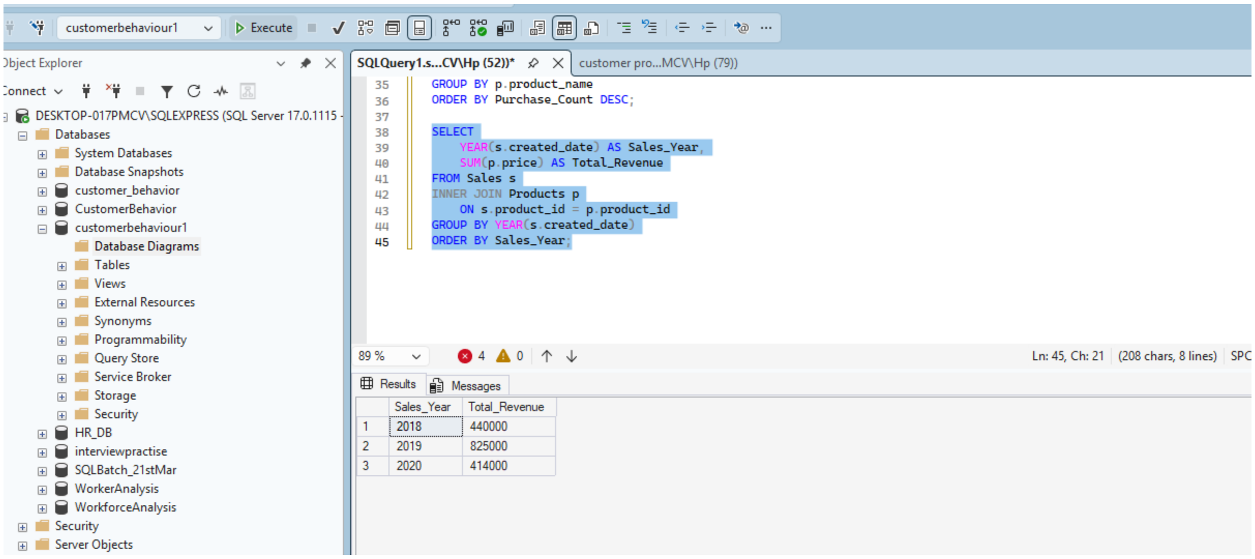
Q. What Is the Most Purchased Product on the Platform?



Key Findings

- Customers show strong preference toward this product.
- The product contributes significantly to overall sales volume.
- The company should ensure adequate stock availability.
- Marketing campaigns can leverage this product to attract customers.

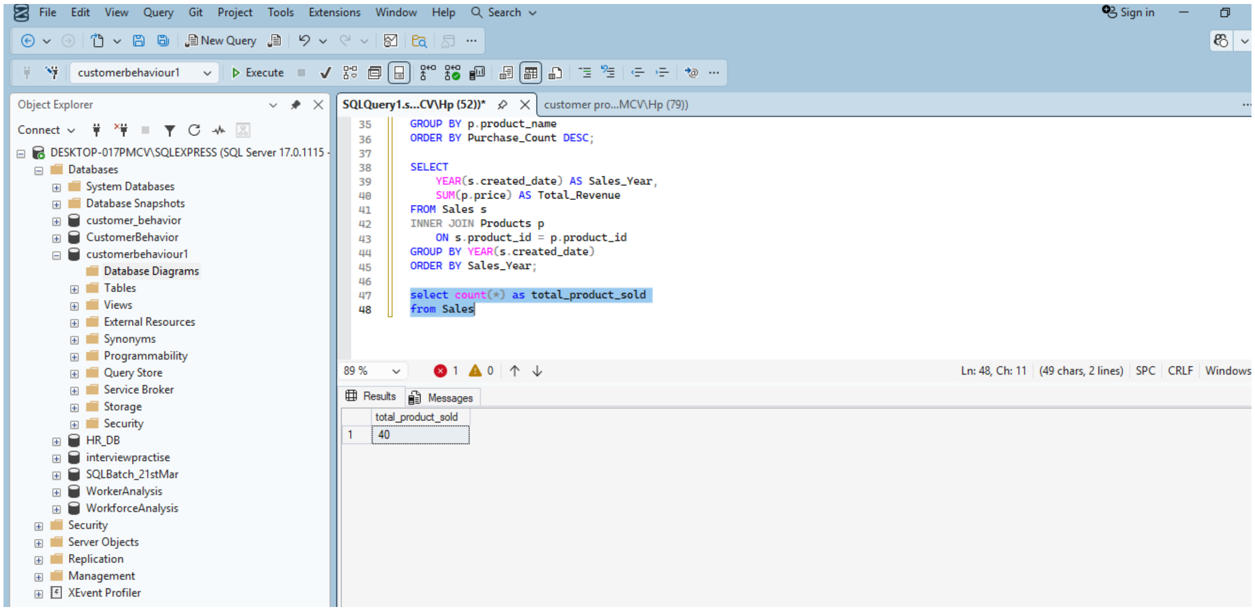
Q. What Is the Total Sales Revenue Generated in Each Year?



Key Findings;-

- The analysis reveals annual sales performance.
- Revenue trends help identify business growth patterns.
- High-performing years indicate strong customer demand.
- Revenue fluctuations may highlight market opportunities or challenges.
- Yearly trend analysis supports future planning and forecasting.

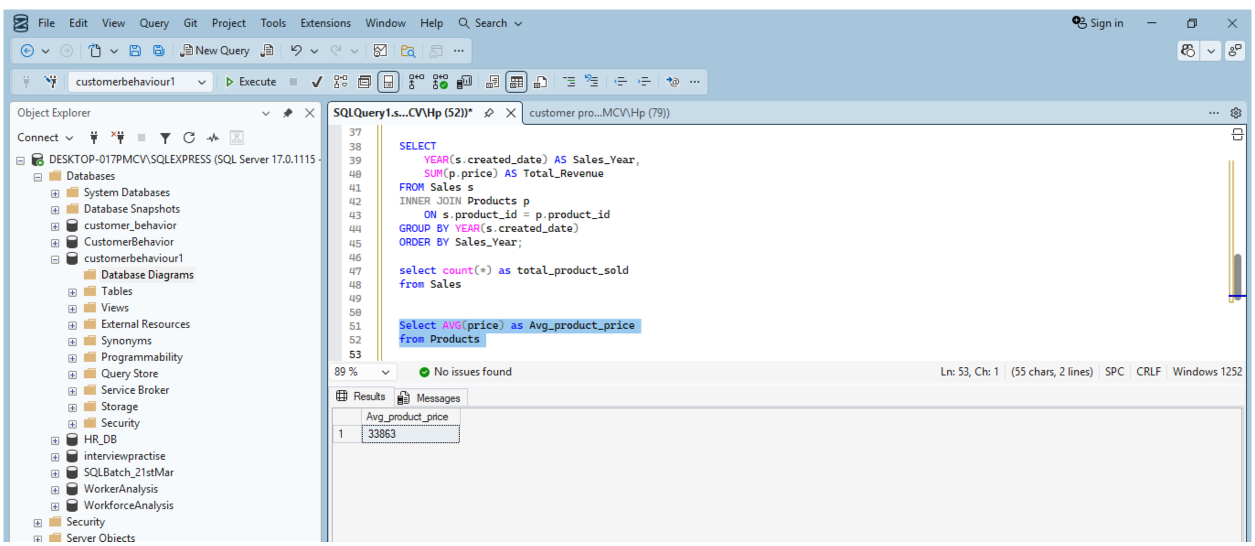
Q. How Many Products Were Sold?



Key Findings;-

- Total products sold on the platform were 40.
- The company has processed a significant number of transactions.
- Sales volume indicates customer activity on the platform.

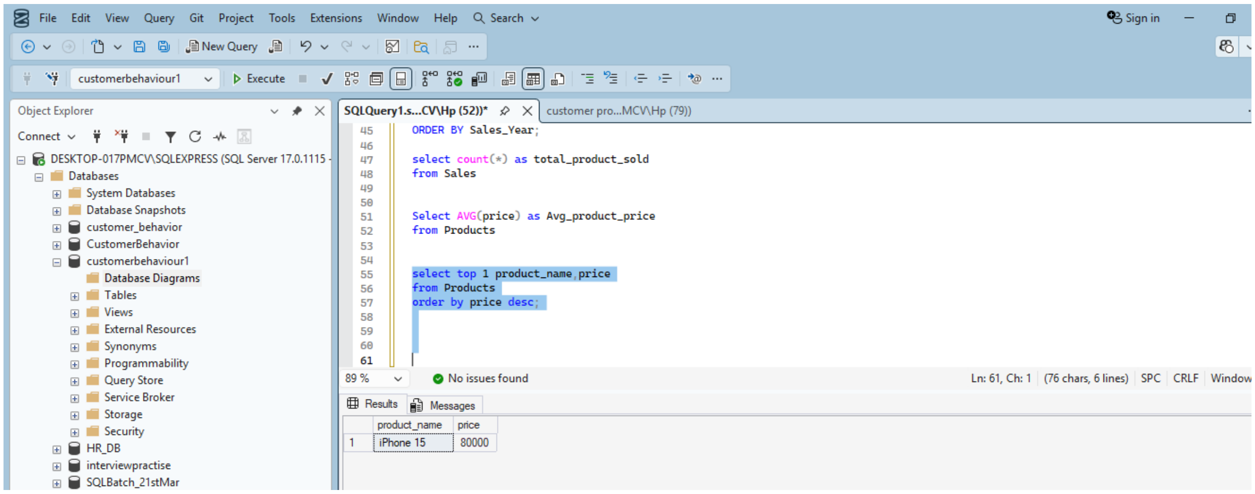
Q. What Is the Average Price of Products?



Key Findings

- The average product price is ₹33863.
- This helps understand the overall pricing strategy.
- The company offers products across different price ranges.

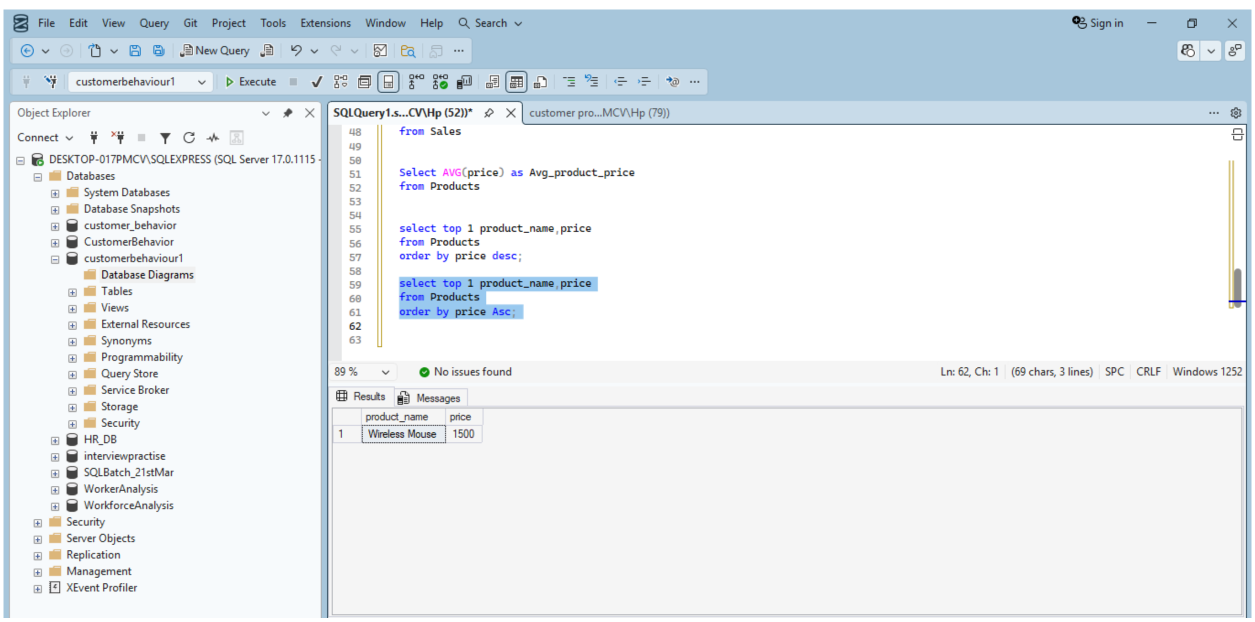
Q. Which Is the Most Expensive Product?



Key Findings;-

- iPhone 15 is the most expensive product.
- Premium products contribute significantly to revenue.
- High-value products attract customers seeking premium electronics.

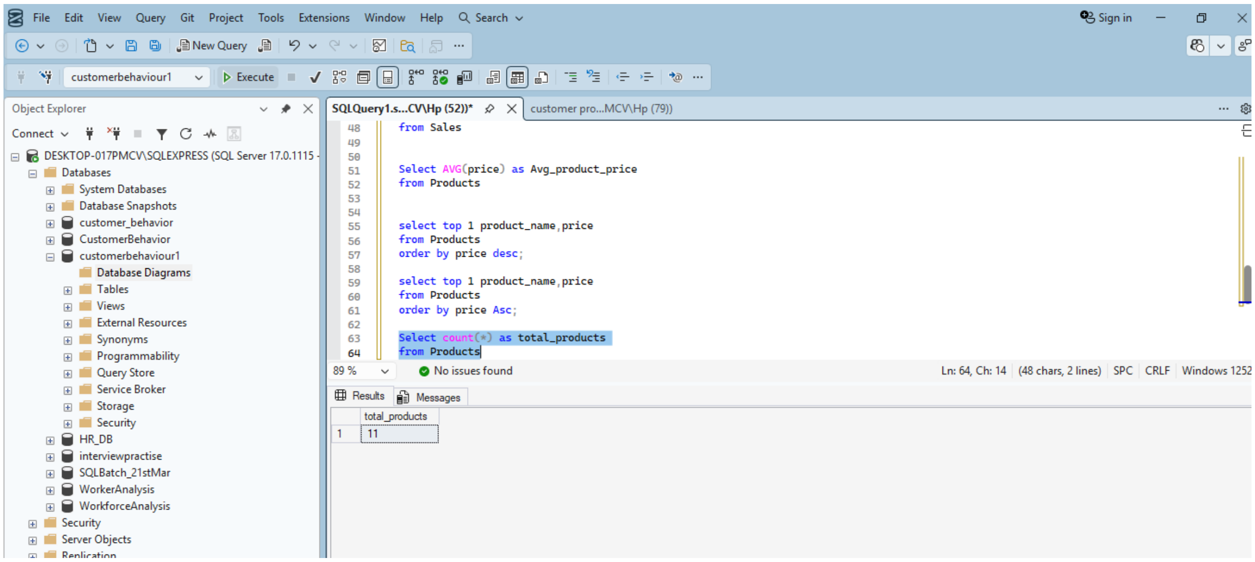
Q. Which Is the Cheapest Product?



Key Findings

- Wireless is the least expensive product.
- Affordable products help attract budget-conscious customers.
- Low-cost products can increase transaction volume.

Q-How Many Products Are Available?



Key Findings

- The store offers 11 products.
- Product variety helps serve different customer needs.
- A diverse catalog improves customer choice.

Conclusion: -

This project successfully analyzed customer purchasing behavior using SQL and provided valuable insights into sales performance, customer engagement, and product popularity. By examining customer transactions, product information, and membership data, meaningful business intelligence was generated.

The analysis identified top-selling products, high-spending customers, purchase frequency patterns, and revenue trends. These insights can help businesses improve decision-making, optimize inventory management, strengthen customer relationships, and increase profitability.

The project also demonstrated the practical application of SQL concepts such as Joins, Aggregate Functions, Group By, Sorting, and Filtering in solving real-world business problems. Overall, this analysis highlights the importance of data-driven decision-making in understanding customer behavior and improving business performance.

Thankyou